

SOLIDWORKS API (.NET VBA C#) — Reference (187/337): IModelDoc2 (part 10 of 17)

SOLIDWORKS API Documentation (converted from CHM)

IModelDoc2 (part 10 of 17) (Type)

IInsertDatumTag2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IInsertDatumTag2 Method (IModelDoc2)

Inserts a datum tag symbol at the selected location.

.NET Syntax

Visual Basic (Declaration)

```
Function IInsertDatumTag2() As DatumTag
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As DatumTag  
  
value = instance.IInsertDatumTag2()
```

C#

```
DatumTag IInsertDatumTag2()
```

C++/CLI

```
DatumTag^ IInsertDatumTag2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Newly created [datum tag](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IInsertDatumTag2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::IInsertDatumTag2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

IInsertGtol Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IInsertGtol Method (IModelDoc2)

Creates a new geometric tolerance symbol (Gtol) in this document.

.NET Syntax

Visual Basic (Declaration)

```
Function IInsertGtol() As Gtol
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As Gtol

value = instance.IInsertGtol()
```

C#

```
Gtol IInsertGtol()
```

C++/CLI

```
Gtol^ IInsertGtol();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Newly created [Gtol](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertGtol](#).

Remarks

The leader attachment points for the newly created Gtol object come from the selections made before calling this method. The initial location of the symbol is near the selection location. If there are no selections, then the Gtol does not have a leader, is free standing, and is initially at the origin of the model or drawing.

This method creates an empty symbol. To fill in the text and symbols of this Gtol, use the pointer returned by this method to access the various get and set methods of the [IGtol](#) interface, such as [IGtol::SetFrameSymbols2](#) and [IGtol::SetFrameValues](#). Use [IGtol::GetAnnotation](#) to retrieve the [IAnnotation](#) object.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::InsertGtol Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertMacroFeature Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertMacroFeature Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertMacroFeature3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IInsertMacroFeature( _
    ByVal CmdFile As System.String, _
    ByVal CmdModule As System.String, _
    ByVal CmdProcedure As System.String, _
    ByVal ParamCount As System.Integer, _
    ByRef ParamNames As System.String, _
    ByRef ParamTypes As System.Integer, _
    ByRef ParamValues As System.String, _
    ByVal PmFile As System.String, _
    ByVal PmModule As System.String, _
    ByVal PmProcedure As System.String _
) As Feature
```

Visual Basic (Usage)

 Copy Code

```

Dim instance As IModelDoc2
Dim CmdFile As System.String
Dim CmdModule As System.String
Dim CmdProcedure As System.String
Dim ParamCount As System.Integer
Dim ParamNames As System.String
Dim ParamTypes As System.Integer
Dim ParamValues As System.String
Dim PmFile As System.String
Dim PmModule As System.String
Dim PmProcedure As System.String
Dim value As Feature

```

```

value = instance.InsertMacroFeature(CmdFile, CmdModule, CmdProcedure, ParamCount, ParamNames,
ParamTypes, ParamValues, PmFile, PmModule, PmProcedure)

```

C#

```

Feature InsertMacroFeature(
    System.string CmdFile,
    System.string CmdModule,
    System.string CmdProcedure,
    System.int ParamCount,
    ref System.string ParamNames,
    ref System.int ParamTypes,
    ref System.string ParamValues,
    System.string PmFile,
    System.string PmModule,
    System.string PmProcedure
)

```

C++/CLI

```

Feature^ InsertMacroFeature(
& System.String^ CmdFile,
& System.String^ CmdModule,
& System.String^ CmdProcedure,
& System.int ParamCount,
& System.String^% ParamNames,
& System.int% ParamTypes,
& System.String^% ParamValues,
& System.String^ PmFile,
& System.String^ PmModule,
& System.String^ PmProcedure
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CmdFile

CmdModule

CmdProcedure

ParamCount

ParamNames

ParamTypes

ParamValues

PmFile

PmModule

PmProcedure

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertMacroFeature](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertMidSurfaceExt Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertMidSurfaceExt Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertMidSurface](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IInsertMidSurfaceExt( _  
    ByVal Placement As System.Double, _  
    ByVal KnitFlag As System.Boolean _  
) As MidSurface2
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Placement As System.Double  
Dim KnitFlag As System.Boolean  
Dim value As MidSurface2
```

```
value = instance.IInsertMidSurfaceExt(Placement, KnitFlag)
```

C#

```
MidSurface2 IInsertMidSurfaceExt(  
    System.double Placement,  
    System.bool KnitFlag  
)
```

C++/CLI

```
MidSurface2^ IInsertMidSurfaceExt(  
& System.double Placement,  
& System.bool KnitFlag  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Placement

Value from -1 to 1 that indicates the desired location of the midsurface relative to the face pair; a value of 0.0 places the midsurface equally between the face pair

KnitFlag

True to sew all the reference surfaces (neutral sheets) into a single sheet (surface) body, false for the midsurface feature to contain individual reference surface geometry

Return Value

[Mid-surface feature](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertMidSurfaceExt](#).

Remarks

A midsurface feature is calculated from the solid body in your part document. This method first finds all parallel pairs of faces from the part solid. For each pair of parallel faces found, it creates a neutral reference surface between the two faces. The placement position of the reference surface is controlled by the placement parameter. This process repeats itself until all parallel body faces are processed. At the end of this process, all the reference surfaces are sewn into a single reference surface if knitFlag is set to True.

The hierarchy of a midsurface feature is:

A midsurface feature contains one or more reference surfaces. If sewn successfully using the knitFlag = True option, then the midsurface feature contains only one reference surface. Each reference surface is considered a sheet body. The sheet body has the normal topology that you would expect to find on a body object (for example, faces, edges, and so on). See the [IMidSurface2](#) object for methods that provide access to this topology.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::InsertMidSurfaceExt Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IInsertNote Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IInsertNote Method (IModelDoc2)

Inserts a note in this document.

.NET Syntax

Visual Basic (Declaration)

```
Function IInsertNote( _  
    ByVal Text As System.String _  
) As Note
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Text As System.String  
Dim value As Note  
  
value = instance.IInsertNote(Text)
```

C#

```
Note IInsertNote(  
    System.string Text  
)
```

C++/CLI

```
Note^ IInsertNote(  
& System.String^ Text  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Text
Text string or symbol for the note (see **Remarks**)

Return Value

Newly created [note](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IInsertNote](#).

Remarks

If you use a symbol in Text, it must be formatted as:

<LibraryName-SymbolName>

where *LibraryName* and *SymbolName* are in the SOLIDWORKS text file **C:\ProgramData\SolidWorks\SolidWorks 20nn\lang\english\gtol.sym**.

NOTE: You must include the right- and left-angle brackets and separate *LibraryName* and *SymbolName* with a hyphen; for example, <MOD-DEG>.

The leader attachment points for the new note come from the selections made before calling this method. The initial location of the note will be near the selection location. If there are no selections, the note:

- Does not have a leader
- Is free standing
- Is at the origin of the model or drawing

This method creates a default note. To adjust the display characteristics of this note, you should use the pointer that is returned by this method to access the properties and get and set methods of the Note object, such as [INote::SetBalloon](#) and [INote::Angle](#). Use [INote::GetAnnotation](#) or [INote::IGetAnnotation](#) to retrieve the [IAnnotation](#) object.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::InsertNote Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IInsertProjectedSketch2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IInsertProjectedSketch2 Method (IModelDoc2)

Projects the selected sketch items from the current sketch onto a selected surface.

.NET Syntax

Visual Basic (Declaration)

```
Function IInsertProjectedSketch2( _  
    ByVal Reverse As System.Integer _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Reverse As System.Integer  
Dim value As Feature
```

```
value = instance.IInsertProjectedSketch2(Reverse)
```

C#

```
Feature IInsertProjectedSketch2(  
    System.int Reverse  
)
```

C++/CLI

```
Feature^ IInsertProjectedSketch2(  
& System.int Reverse  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Reverse

1 to reverse the projected direction

Return Value

Newly created [feature](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertProjectedSketch2](#).

Remarks

You can reverse the direction in which the curve is projected. This is necessary only when the selected face wraps around the plane of the curve. For example, if the sketch being projected is surrounded by a cylindrical face, then two possible projections exist. The Reverse argument switches the direction based on the normal vector of the sketch. The default direction is along the sketch normal.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::InsertProjectedSketch Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSheetMetalEdgeFlange Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertSheetMetalEdgeFlange Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertSheetMetalEdgeFlange2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IInsertSheetMetalEdgeFlange( _  
    ByVal FlangeEdge As Edge, _  
    ByVal SketchFeat As Feature, _  
    ByVal BooleanOptions As System.Integer, _  
    ByVal DAngle As System.Double, _  
    ByVal DRadius As System.Double, _  
    ByVal BendPosition As System.Integer, _  
    ByVal DOffsetDist As System.Double, _  
    ByVal ReliefType As System.Integer, _  
    ByVal DReliefRatio As System.Double, _  
    ByVal DReliefWidth As System.Double, _  
    ByVal DReliefDepth As System.Double _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim FlangeEdge As Edge  
Dim SketchFeat As Feature  
Dim BooleanOptions As System.Integer  
Dim DAngle As System.Double  
Dim DRadius As System.Double  
Dim BendPosition As System.Integer  
Dim DOffsetDist As System.Double  
Dim ReliefType As System.Integer  
Dim DReliefRatio As System.Double  
Dim DReliefWidth As System.Double  
Dim DReliefDepth As System.Double  
Dim value As Feature
```

```
value = instance.IInsertSheetMetalEdgeFlange(FlangeEdge, SketchFeat, BooleanOptions, DAngle,  
DRadius, BendPosition, DOffsetDist, ReliefType, DReliefRatio, DReliefWidth, DReliefDepth)
```

C#

```

Feature IInsertSheetMetalEdgeFlange(
    Edge FlangeEdge,
    Feature SketchFeat,
    System.int BooleanOptions,
    System.double DAngle,
    System.double DRadius,
    System.int BendPosition,
    System.double DOffsetDist,
    System.int ReliefType,
    System.double DReliefRatio,
    System.double DReliefWidth,
    System.double DReliefDepth
)

```

C++/CLI

```

Feature^ IInsertSheetMetalEdgeFlange(
    & Edge^ FlangeEdge,
    & Feature^ SketchFeat,
    & System.int BooleanOptions,
    & System.double DAngle,
    & System.double DRadius,
    & System.int BendPosition,
    & System.double DOffsetDist,
    & System.int ReliefType,
    & System.double DReliefRatio,
    & System.double DReliefWidth,
    & System.double DReliefDepth
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FlangeEdge

SketchFeat

BooleanOptions

DAngle

DRadius

BendPosition

DOffsetDist

ReliefType

DReliefRatio

DReliefWidth

DReliefDepth

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSheetMetalEdgeFlange](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSketchForEdgeFlange Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertSketchForEdgeFlange Method (IModelDoc2)

Inserts a sketch for [IFeatureManager::InsertSheetMetalEdgeFlange2](#) in this sheet metal part.

.NET Syntax

```
Function IInsertSketchForEdgeFlange( _
    ByVal FlangeEdge As Edge, _
    ByVal DAngle As System.Double, _
    ByVal FlipDir As System.Boolean _
) As Feature
```



```
Dim instance As IModelDoc2
Dim FlangeEdge As Edge
Dim DAngle As System.Double
Dim FlipDir As System.Boolean
Dim value As Feature

value = instance.IInsertSketchForEdgeFlange(FlangeEdge, DAngle, FlipDir)
```

```
Feature IInsertSketchForEdgeFlange(
    Edge FlangeEdge,
    System.double DAngle,
    System.bool FlipDir
)
```

```
Feature^ IInsertSketchForEdgeFlange(
    & Edge^ FlangeEdge,
    & System.double DAngle,
    & System.bool FlipDir
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FlangeEdge

Edge to which to apply flange

DAngle

Angle of flange

FlipDir

True reverses the offset direction of the flange, false does not

Return Value

Sketch for the edge flange, returned as a [feature](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSketchForEdgeFlange](#).

Remarks

After calling this method, you must create the profile for the flange on the appropriate plane. After creating the profile, call [IFeatureManager::InsertSheetMetalEdgeFlange2](#) to create the flange.

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

[IModelDoc2::InsertSheetMetalMiterFlange Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : [InsertSketchText Method \(IModelDoc2\)](#)

Obsolete. Superseded by [IModelDoc2::InsertSketchText](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IInsertSketchText( _  
    ByVal Ptx As System.Double, _  
    ByVal Pty As System.Double, _  
    ByVal Ptz As System.Double, _  
    ByVal Text As System.String, _  
    ByVal Alignment As System.Integer, _  
    ByVal FlipDirection As System.Integer, _  
    ByVal HorizontalMirror As System.Integer, _  
    ByVal WidthFactor As System.Integer, _  
    ByVal SpaceBetweenChars As System.Integer _  
) As SketchText
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Ptx As System.Double  
Dim Pty As System.Double  
Dim Ptz As System.Double  
Dim Text As System.String  
Dim Alignment As System.Integer  
Dim FlipDirection As System.Integer  
Dim HorizontalMirror As System.Integer  
Dim WidthFactor As System.Integer  
Dim SpaceBetweenChars As System.Integer  
Dim value As SketchText  
  
value = instance.IInsertSketchText(Ptx, Pty, Ptz, Text, Alignment, FlipDirection,  
HorizontalMirror, WidthFactor, SpaceBetweenChars)
```

C#

```
SketchText IInsertSketchText(  
    System.double Ptx,  
    System.double Pty,  
    System.double Ptz,  
    System.string Text,  
    System.int Alignment,  
    System.int FlipDirection,  
    System.int HorizontalMirror,  
    System.int WidthFactor,  
    System.int SpaceBetweenChars  
)
```

C++/CLI

```
SketchText^ IInsertSketchText(  
& System.double Ptx,  
& System.double Pty,  
& System.double Ptz,  
& System.String^ Text,  
& System.int Alignment,  
& System.int FlipDirection,  
& System.int HorizontalMirror,  
& System.int WidthFactor,  
& System.int SpaceBetweenChars  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Ptx
X coordinate of text

Pty
Y coordinate of text

Ptz
Z coordinate of text

Text
Text to insert (see **Remarks**)

Alignment
Alignment:

- 0 = Left
- 1 = Center
- 2 = Right
- 3 = Fully justified

(see **Remarks**)

FlipDirection
1 to flip text vertically about the selected entity, 0 to not (see **Remarks**)

HorizontalMirror
1 to flip text horizontally, 0 to not

WidthFactor
6 <= Percentage by which to evenly widen each character in the text block <= 1667

SpaceBetweenChars
1 <= Percentage of space between each character in the text block <= 10000; valid only if Alignment != 3

Return Value

[Sketch text](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertSketchText](#).

Remarks

Alignment and FlipDirection are valid only when a curve, edge, or sketch segment is selected. Text appears along the selected entity. If an entity is not selected, Text appears horizontally starting at the origin. [Select](#) the curve, edge, or sketch segment with Mark = 1.

See the SOLIDWORKS Design user-interface help > **Sketching** > **Sketch Entities** > **Sketch Text** > **SketchText PropertyManager** topic for more information about this functionality.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::DissolveSketchText Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IInsertWeldSymbol3 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IInsertWeldSymbol3 Method (IModelDoc2)

Inserts a weld symbol into the model.

.NET Syntax

Visual Basic (Declaration)

```
Function IInsertWeldSymbol3() As WeldSymbol
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As WeldSymbol

value = instance.IInsertWeldSymbol3()
```

C#

```
WeldSymbol IInsertWeldSymbol3()
```

C++/CLI

```
WeldSymbol^ IInsertWeldSymbol3();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Newly created [weld symbol](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IInsertWeldSymbol3](#).

Remarks

To use this method, insert the weld symbol into the model and then manipulate the properties and methods on the [IWeldSymbol](#) object.

A list of weld symbol names can be found in **C:\ProgramData\SolidWorks\SolidWorks 20nn\lang\english\lgto1.sym**. The currently supported list of ISO weld symbols is:

- BUTT
- BUSQ
- BUSV
- BUSB
- BUSVBR
- BUSBR
- BUSU
- BUSJ
- BACK
- FILL
- PLUG
- SPOT
- SEAM
- SEAMC
- JSPT
- JSM

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::IInsertWeldSymbol3 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

Gets and sets the light source property values.

Visual Basic Syntax

Visual Basic (Declaration)

```
Property ILightSourcePropertyValues( _  
    ByVal ID As System.Integer _  
) As System.Double
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
Dim ID As System.Integer  
Dim value As System.Double  
  
instance.ILightSourcePropertyValues(ID) = value  
  
value = instance.ILightSourcePropertyValues(ID)
```

C#

```
System.double ILightSourcePropertyValues(  
    System.int ID  
) {get; set;}
```

C++/CLI

```
property System.double ILightSourcePropertyValues {  
    System.double get(System.int ID);  
    void set (System.int ID, System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ID
Light source ID

Property Value

- in-process, unmanaged C++: Pointer to an array of 19 doubles containing the light source properties (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ILightSourcePropertyValues](#).

Remarks

The light source ID ranges from 0 to n , where n = (the total number of light sources - 1). To get the total number of light sources, use [IModelDoc2::GetLightSourceCount](#).

The return values for this property is the following array of 19 doubles:

[*type*, *Intensity*, *Color*[3], *Position*[3], *Direction*[3], *coneAngle*, *falloff*[3], *Ambient*, *Specular*, *isDisabled*, *SpotExponent*]

where:

- type* = an integer packed into a double that represents the light source type where valid types are taken from OpenGL definitions (LIGHT_EYE, LIGHT_AMBIENT, LIGHT_SPOT, LIGHT_POINT, LIGHT_DISTANT)
- Intensity* = Light-source intensity (diffuseness) where values range from 0 to 1.
- Color*[3] = Light-source color in the form of an array of 3 doubles (R, G, B) where values range from 0 to 1.
- Position*[3] = Light-source position for spot lights in the form of an array of 3 doubles (X, Y, Z) in model space.
- Direction*[3] = Light-source direction in the form of an array of 3 doubles (i, j, k)
- coneAngle* = Light-source cone half angle in degrees where valid values are from 0 to 90 and 180.
- falloff*[3] = Light-source falloff in the form of an array of 3 doubles (Kc, Kl, Kq). These values result in changes in light intensity as the distance between the light position and the vertex change.

NOTE: The falloff[3] values are not supported in SOLIDWORKS 2011 and later.

The result is driven by the following equation:
$$\left[1 / (Kc + D * KI + D * D * Kq) \right]$$

where:

D = the distance between the light's position and the vertex

Kc = constant attenuation

KI = linear attenuation

Kq = quadratic attenuation

- *Ambient* = light-source ambient Intensity
- *Specular* = light-source specular Intensity
- *isDisabled* = an integer packed into a double. True means that the light is disabled and false means the light is not disabled.
- *SpotExponent* = spot exponent

NOTE: The SpotExponent value is not supported in SOLIDWORKS 2011 and later.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::LightSourcePropertyValues Property](#)
[IModelDoc2::LightSourceUserName Property](#)
[IModelDoc2::AddLightSource Method](#)
[IModelDoc2::AddLightSourceExtProperty Method](#)
[IModelDoc2::DeleteLightSource Method](#)
[IModelDoc2::GetLightSourceExtProperty Method](#)
[IModelDoc2::GetLightSourceIdFromName Method](#)
[IModelDoc2::GetLightSourceName Method](#)
[IModelDoc2::ResetLightSourceExtProperty Method](#)
[IModelDoc2::SetLightSourceName Method](#)
[IModelDoc2::SetLightSourcePropertyValuesVB Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IListAuxiliaryExternalFileReferences Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IListAuxiliaryExternalFileReferences Method (IModelDoc2)

Gets the names of auxiliary external file references for this model.

.NET Syntax

Visual Basic (Declaration)

```
Sub IListAuxiliaryExternalFileReferences( _  
    ByVal NumRefs As System.Integer, _  
    ByRef Feature As System.String, _  
    ByRef ExternalFileName As System.String _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim NumRefs As System.Integer  
Dim Feature As System.String  
Dim ExternalFileName As System.String  
  
instance.IListAuxiliaryExternalFileReferences(NumRefs, Feature, ExternalFileName)
```

C#

```
void IListAuxiliaryExternalFileReferences(
    System.int NumRefs,
    out System.string Feature,
    out System.string ExternalFileName
)
```

C++/CLI

```
void IListAuxiliaryExternalFileReferences(
    & System.int NumRefs,
    & [Out] System.String^ Feature,
    & [Out] System.String^ ExternalFileName
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumRefs

Number of external references

Feature

Array of size NumRefs containing the names of features that include auxiliary external references

ExternalFileName

Array of size NumRefs containing the names of the auxiliary external files

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IListAuxiliaryExternalFileReferences](#).

Remarks

Call [IModelDoc2::ListAuxiliaryExternalFileReferencesCount](#) before calling of this method to determine the size of the arrays.

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

[IModelDoc2::ListAuxiliaryExternalFileReferences Method](#)

[IModelDocExtension::IListExternalFileReferences Method](#)

[IModelDocExtension::ListExternalFileReferences Method](#)

[IModelDoc2::ListAuxiliaryExternalFileReferencesCount Method](#)

[IModelDoc2::LockAllExternalReferences Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IListExternalFileReferences Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : [IListExternalFileReferences Method \(IModelDoc2\)](#)

Obsolete. Superseded by [IModelDocExtension::ListExternalReferences](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IListExternalFileReferences( _
    ByVal UseSearchRules As System.Boolean, _
    ByVal NumRefs As System.Integer _
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim UseSearchRules As System.Boolean
Dim NumRefs As System.Integer
```

```
Dim value As System.String
```

```
value = instance.IListExternalFileReferences(UseSearchRules, NumRefs)
```

C#

```
System.string IListExternalFileReferences(  
    System.bool UseSearchRules,  
    System.int NumRefs  
)
```

C++/CLI

```
System.String^ IListExternalFileReferences(  
& System.bool UseSearchRules,  
& System.int NumRefs  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UseSearchRules

NumRefs

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IListExternalFileReferences](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IListExternalFileReferences2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IListExternalFileReferences2 Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::ListExternalReferences](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub IListExternalFileReferences2( _  
    ByVal NumRefs As System.Integer, _  
    ByRef ModelPathName As System.String, _  
    ByRef CompPathName As System.String, _  
    ByRef Feature As System.String, _  
    ByRef DataType As System.String, _  
    ByRef Status As System.Integer, _  
    ByRef RefEntity As System.String, _  
    ByRef FeatComp As System.String _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim NumRefs As System.Integer  
Dim ModelPathName As System.String  
Dim CompPathName As System.String  
Dim Feature As System.String  
Dim DataType As System.String  
Dim Status As System.Integer  
Dim RefEntity As System.String  
Dim FeatComp As System.String
```

```
instance.IListExternalFileReferences2(NumRefs, ModelPathName, CompPathName, Feature, DataType,
Status, RefEntity, FeatComp)
```

C#

```
void IListExternalFileReferences2(
    System.int NumRefs,
    out System.string ModelPathName,
    out System.string CompPathName,
    out System.string Feature,
    out System.string DataType,
    out System.int Status,
    out System.string RefEntity,
    out System.string FeatComp
)
```

C++/CLI

```
void IListExternalFileReferences2(
    & System.int NumRefs,
    & [Out] System.String^ ModelPathName,
    & [Out] System.String^ CompPathName,
    & [Out] System.String^ Feature,
    & [Out] System.String^ DataType,
    & [Out] System.int Status,
    & [Out] System.String^ RefEntity,
    & [Out] System.String^ FeatComp
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

- NumRefs
- ModelPathName
- CompPathName
- Feature
- DataType
- Status
- RefEntity
- FeatComp

 Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IListExternalFileReferences2](#).

 See Also

- [IModelDoc2 Interface](#)
- [IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IMaterialPropertyValues Property (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IMaterialPropertyValues Property (IModelDoc2)

Gets or sets a material's properties in the active configuration.

 .NET Syntax

Visual Basic (Declaration)

Property IMaterialPropertyValues As System.Double

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Double

instance.IMaterialPropertyValues = value
value = instance.IMaterialPropertyValues
```

C#

```
System.double IMaterialPropertyValues {get; set;}
```

C++/CLI

```
property System.double IMaterialPropertyValues {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of 9 doubles describing the material's values (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IMaterialPropertyValues](#).

Remarks

The format of the return values is an array of doubles:

[*R, G, B, Ambient, Diffuse, Specular, Shininess, Transparency, Emission*]

To see a color change, you must:

1. Specify *R*, *G*, and *B*, each with a value between 0.0 and 1.0, inclusive. (These values are internally multiplied by 255.0 to determine the RGB color.)
2. Specify the reflectivity properties (*Diffuse*, *Specular*, *Shininess* (1.0 - Specular spread/Blurriness)), each with a value greater than 0.0 and less than or equal to 1.0.
3. Specify *Ambient*, *Transparency* and *Emission*, each with a value between 0.0 and 1.0, inclusive.
4. Refresh the graphics area after setting this property.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDocExtension::GetMaterialPropertyValues Method](#)
[IModelDocExtension::!GetMaterialPropertyValues Method](#)
[IModelDocExtension::!RemoveMaterialProperty Method](#)
[IModelDocExtension::RemoveMaterialProperty Method](#)
[IModelDocExtension::SetMaterialPropertyValues Method](#)
[IModelDocExtension::!SetMaterialPropertyValues Method](#)
[IModelDoc2::MaterialPropertyValues Property](#)
[IModelDoc2::MaterialUserName Property](#)
[IModelDoc2::MaterialIdName Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IMultiSelectByRay Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : IMultiSelectByRay Method (IModelDoc2)

Selects multiple objects of the specified type that are intersected by a ray from point (x,y,z in meters) in direction vector (x,y,z) within a distance radius.

.NET Syntax

Visual Basic (Declaration)

```
Function IMultiSelectByRay( _
    ByRef PointIn As System.Double, _
    ByRef VectorIn As System.Double, _
```

```

    ByVal RadiusIn As System.Double, _
    ByVal TypeWanted As System.Integer, _
    ByVal Append As System.Boolean _
) As System.Boolean

```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim PointIn As System.Double
Dim VectorIn As System.Double
Dim RadiusIn As System.Double
Dim TypeWanted As System.Integer
Dim Append As System.Boolean
Dim value As System.Boolean

value = instance.IMultiSelectByRay(PointIn, VectorIn, RadiusIn, TypeWanted, Append)

```

C#

```

System.bool IMultiSelectByRay(
    ref System.double PointIn,
    ref System.double VectorIn,
    System.double RadiusIn,
    System.int TypeWanted,
    System.bool Append
)

```

C++/CLI

```

System.bool IMultiSelectByRay(
    & System.double% PointIn,
    & System.double% VectorIn,
    & System.double RadiusIn,
    & System.int TypeWanted,
    & System.bool Append
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PointIn

Array containing 3 doubles that define the start point of the ray

VectorIn

Array containing 3 doubles that define the direction of the ray

RadiusIn

Radius of the ray in meters

TypeWanted

Type of objects to select as defined in [swSelectType_e](#)

Append

True to append the selections to the current selection list, false to not

Return Value

True if an object is selected, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::IMultiSelectByRay](#).

Remarks

This method:

- Defines a cylindrical region of infinite length that begins at PointIn, runs parallel to VectorIn, and has a radius of RadiusIn. If you specified edges or vertices, then any edge or vertex found within the selection cylinder is selected.
- Selects only entity objects, which include faces, edges, and vertices. You cannot use this function to select sketches. When selecting face entities, RadiusIn is ignored and a cylinder of infinitely small radius is used.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::MultiSelectByRay Method](#)
[IModelDoc2::SelectByRay Method](#)
[IModelDoc2::GetRayIntersectionsPoints Method](#)
[IModelDoc2::GetRayIntersectionsTopology Method](#)
[IModelDoc2::IGetRayIntersectionsPoints Method](#)
[IModelDoc2::IGetRayIntersectionsTopology Method](#)
[IModelDoc2::IRayIntersections Method](#)
[IModelDoc2::ISelectByRay Method](#)
[IModelDoc2::RayIntersections Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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Insert3DSketch Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : Insert3DSketch Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::Insert3DSketch2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub Insert3DSketch()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
```

```
instance.Insert3DSketch()
```

C#

```
void Insert3DSketch()
```

C++/CLI

```
void Insert3DSketch();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::Insert3DSketch](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

Insert3DSketch2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : Insert3DSketch2 Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::Insert3DSketch](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub Insert3DSketch2( _  
    ByVal UpdateEditRebuild As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim UpdateEditRebuild As System.Boolean

instance.Insert3DSketch2(UpdateEditRebuild)
```

C#

```
void Insert3DSketch2(
    System.bool UpdateEditRebuild
)
```

C++/CLI

```
void Insert3DSketch2(
    & System.bool UpdateEditRebuild
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UpdateEditRebuild

True if you want to edit and rebuild the document, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::Insert3DSketch2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[ISketch::Is3D Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Insert3DSplineCurve Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : Insert3DSplineCurve Method (IModelDoc2)

Inserts a 3D-spline curve through the selected reference points.

.NET Syntax

Visual Basic (Declaration)

```
Sub Insert3DSplineCurve( _
    ByVal CurveClosed As System.Boolean _
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim CurveClosed As System.Boolean

instance.Insert3DSplineCurve(CurveClosed)
```

C#

```
void Insert3DSplineCurve(
    System.bool CurveClosed
)
```

C++/CLI

```
void Insert3DSplineCurve(  
& System.bool CurveClosed  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CurveClosed

True if you want the curve to be closed, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::Insert3DSplineCurve](#).

Example

[Insert a 3D-Spline Curve \(VBA\)](#)

[Create Curve Through Reference Points \(VBA\)](#)

[Create Curve Through Reference Points \(VB.NET\)](#)

[Create Curve Through Reference Points \(C#\)](#)

Remarks

Before calling this method, select the reference points by calling [IModelDocExtension::SelectbyID2](#) with Marks of 1.

To create 2D splines on a sketch, use [IModelDoc2::SketchSpline](#) or [IModelDoc2::CreateSpline](#) or [IModelDoc2::ICreateSpline](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

Availability

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SOLIDWORKS API Help

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InsertAxis Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertAxis Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::InsertAxis2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertAxis() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
```

```
Dim value As System.Boolean
```

```
value = instance.InsertAxis()
```

C#

```
System.bool InsertAxis()
```

C++/CLI

```
System.bool InsertAxis();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertAxis](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

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InsertAxis2 Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertAxis2 Method (IModelDoc2)

Inserts a reference axis based on the currently selected items with an option to automatically size the axis.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertAxis2( _  
    ByVal AutoSize As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim AutoSize As System.Boolean  
Dim value As System.Boolean  
  
value = instance.InsertAxis2(AutoSize)
```

C#

```
System.bool InsertAxis2(  
    System.bool AutoSize  
)
```

C++/CLI

```
System.bool InsertAxis2(  
& System.bool AutoSize  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AutoSize

True if axis is to be automatically sized, false if not

Return Value

True if the reference axis is created successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertAxis2](#).

Example

[Create Revolve Features \(VBA\)](#)
[Create Revolve Features \(VB.NET\)](#)
[Create Revolve Features \(C#\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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InsertBendTableEdit Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertBendTableEdit Method (IModelDoc2)

Inserts a bend table and puts the bend table into its edit state.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertBendTableEdit()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.InsertBendTableEdit()
```

C#

```
void InsertBendTableEdit()
```

C++/CLI

```
void InsertBendTableEdit();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertBendTableEdit](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::InsertBendTableNew Method](#)
[IModelDoc2::InsertBendTableOpen Method](#)
[IModelDoc2::DeleteBendTable Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertBendTableNew Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertBendTableNew Method (IModelDoc2)

Inserts a new bend table into the model document.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertBendTableNew( _  
    ByVal FileName As System.String, _  
    ByVal Units As System.String, _  
    ByVal Type As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim FileName As System.String
Dim Units As System.String
Dim Type As System.String
Dim value As System.Boolean

value = instance.InsertBendTableNew(FileName, Units, Type)

```

C#

```

System.bool InsertBendTableNew(
    System.string FileName,
    System.string Units,
    System.string Type
)

```

C++/CLI

```

System.bool InsertBendTableNew(
    & System.String^ FileName,
    & System.String^ Units,
    & System.String^ Type
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

File name of this new bend table

Units

- Millimeters
- Centimeters
- Meters
- Inches
- Feet

Type

- Bend Allowance
- Bend Deduction

Return Value

True if the new table is successfully inserted, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertBendTableNew](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::DeleteBendTable Method](#)
[IModelDoc2::InsertBendTableEdit Method](#)
[IModelDoc2::InsertBendTableOpen Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertBendTableOpen Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertBendTableOpen Method (IModelDoc2)

Inserts an existing bend table from a file into this model document.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertBendTableOpen( _  
    ByVal FileName As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim FileName As System.String  
Dim value As System.Boolean  
  
value = instance.InsertBendTableOpen(FileName)
```

C#

```
System.bool InsertBendTableOpen(  
    System.string FileName  
)
```

C++/CLI

```
System.bool InsertBendTableOpen(  
& System.String^ FileName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

File name of the bend table to insert into this model document

Return Value

True if the table is successfully inserted, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertBendTableOpen](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::InsertBendTableEdit Method](#)
[IModelDoc2::InsertBendTableNew Method](#)
[IModelDoc2::DeleteBendTable Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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InsertBkgImage Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertBkgImage Method (IModelDoc2)

Inserts the scene background image.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertBkgImage( _  
    ByVal NewName As System.String _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim NewName As System.String

instance.InsertBkgImage(NewName)
```

C#

```
void InsertBkgImage(
    System.string NewName
)
```

C++/CLI

```
void InsertBkgImage(
    & System.String^ NewName
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NewName
Path to image file including file extension

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertBkgImage](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::DeleteBkgImage Method](#)
[IModelDoc2::GetSceneBkgDIB Method](#)
[IModelDoc2::SceneBkgImageFileName Property](#)
[IModelDoc2::SetSceneBkgDIB Method](#)
[IModelDocExtension::SetSceneBkgDIBx64 Method](#)
[IModelDocExtension::GetSceneBkgDIBx64 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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InsertBOMBalloon Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertBOMBalloon Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::InsertBOMBalloon2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertBOMBalloon()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2

instance.InsertBOMBalloon()
```

C#

```
void InsertBOMBalloon()
```

C++/CLI

```
void InsertBOMBalloon();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertBOMBalloon](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertBOMBalloon2 Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertBOMBalloon2 Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::InsertBOMBalloon](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertBOMBalloon2( _  
    ByVal Style As System.Integer, _  
    ByVal Size As System.Integer, _  
    ByVal UpperTextStyle As System.Integer, _  
    ByVal UpperText As System.String, _  
    ByVal LowerTextStyle As System.Integer, _  
    ByVal LowerText As System.String _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Style As System.Integer  
Dim Size As System.Integer  
Dim UpperTextStyle As System.Integer  
Dim UpperText As System.String  
Dim LowerTextStyle As System.Integer  
Dim LowerText As System.String  
Dim value As System.Object
```

```
value = instance.InsertBOMBalloon2(Style, Size, UpperTextStyle, UpperText, LowerTextStyle,  
LowerText)
```

C#

```
System.object InsertBOMBalloon2(  
    System.int Style,  
    System.int Size,  
    System.int UpperTextStyle,  
    System.string UpperText,  
    System.int LowerTextStyle,  
    System.string LowerText  
)
```

C++/CLI

```
System.Object^ InsertBOMBalloon2(  
& System.int Style,  
& System.int Size,  
& System.int UpperTextStyle,  
& System.String^ UpperText,  
& System.int LowerTextStyle,  
& System.String^ LowerText  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Style

Balloon style as defined in [swBalloonStyle_e](#)

Size

Balloon size as defined in [swBalloonFit_e](#)

UpperTextStyle

Text style for the upper text of the balloon as defined in [swBalloonTextContent_e](#)

UpperText

Text string to be placed in the upper text of the balloon

LowerTextStyle

Text style for the lower text of the balloon as defined in [swBalloonTextContent_e](#)

LowerText

Text string to be placed in the lower text of the balloon

Return Value

Newly created [note](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertBOMBalloon2](#).

Example

[Insert BOM Balloon \(VBA\)](#)

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

[IModelDoc2::InsertBOMBalloon2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertCompositeCurve Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertCompositeCurve Method (IModelDoc2)

Inserts a composite curve based on selections.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertCompositeCurve() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Boolean
```

```
value = instance.InsertCompositeCurve()
```

C#

```
System.bool InsertCompositeCurve()
```

C++/CLI

```
System.bool InsertCompositeCurve();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertCompositeCurve](#).

Example

[Insert a Composite Curve \(C#\)](#)
[Insert a Composite Curve \(VB.NET\)](#)
[Insert a Composite Curve \(VBA\)](#)

Remarks

To use this method, select the entities by calling [IModelDocExtension::SelectByID2](#) with a mark number of 1.
See the SOLIDWORKS Help for information about which entities are valid for selection.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertConnectionPoint Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertConnectionPoint Method (IModelDoc2)

Adds a connection point based on the selected point and selected planar item.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertConnectionPoint()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.InsertConnectionPoint()
```

C#

```
void InsertConnectionPoint()
```

C++/CLI

```
void InsertConnectionPoint();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertConnectionPoint](#).

Remarks

If the selection set is not complete, then the **Insert Connection Point** dialog is displayed.
The connection point is the point on the fitting that defines where the connection to other pipe items begins or ends.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertCoordinateSystem Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertCoordinateSystem Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertCoordinateSystem](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertCoordinateSystem( _  
    ByVal XFlippedIn As System.Boolean, _  
    ByVal YFlippedIn As System.Boolean, _  
    ByVal ZFlippedIn As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim XFlippedIn As System.Boolean  
Dim YFlippedIn As System.Boolean  
Dim ZFlippedIn As System.Boolean  
Dim value As System.Boolean  
  
value = instance.InsertCoordinateSystem(XFlippedIn, YFlippedIn, ZFlippedIn)
```

C#

```
System.bool InsertCoordinateSystem(  
    System.bool XFlippedIn,  
    System.bool YFlippedIn,  
    System.bool ZFlippedIn  
)
```

C++/CLI

```
System.bool InsertCoordinateSystem(  
& System.bool XFlippedIn,  
& System.bool YFlippedIn,  
& System.bool ZFlippedIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XFlippedIn

True to flip the x direction, false to not

YFlippedIn

True to flip the y direction, false to not

ZFlippedIn

True to flip the z direction, false to not

Return Value

True if coordinate system is added successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertCoordinateSystem](#).

Example

[Insert Coordinate System at Center of Mass \(VBA\)](#)

Remarks

Make the selections using [IModelDocExtension::SelectByID2](#) with a mark of:

- 1 - Origin
- 2 - X axis
- 4 - Y axis
- 8 - Z axis

This method:

- does not require all three axis to be selected. The behavior is the same as interactively creating a coordinate system. See the SOLIDWORKS Help for more information.
- works in section-view mode, but not if temporary geometry that only exists in section-view mode is selected.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetCurrentCoordinateSystemName Method](#)
[IModelDocExtension::GetCoordinateSystemTransformByName Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertCosmeticThread Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertCosmeticThread Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertCosmeticThread2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertCosmeticThread( _  
    ByVal Type As System.Short, _  
    ByVal Depth As System.Double, _  
    ByVal Length As System.Double, _  
    ByVal Note As System.String _  
)
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
Dim Type As System.Short  
Dim Depth As System.Double  
Dim Length As System.Double  
Dim Note As System.String  
  
instance.InsertCosmeticThread(Type, Depth, Length, Note)
```

C#

```
void InsertCosmeticThread(  
    System.short Type,  
    System.double Depth,  
    System.double Length,  
    System.string Note  
)
```

C++/CLI

```
void InsertCosmeticThread(  
& System.short Type,  
& System.double Depth,  
& System.double Length,
```

```
& System.String^ Note
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type

Depth

Length

Note

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertCosmeticThread](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertCurveFile Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertCurveFile Method (IModelDoc2)

Creates a curve.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertCurveFile( _
    ByVal FileName As System.String _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim FileName As System.String
Dim value As System.Boolean

value = instance.InsertCurveFile(FileName)
```

C#

```
System.bool InsertCurveFile(
    System.string FileName
)
```

C++/CLI

```
System.bool InsertCurveFile(
    & System.String^ FileName
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Path and file name containing the point data

Return Value

True if the curve is created successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertCurveFile](#).

Remarks

The curve goes through the points in the specified file. The points in the specified input file can have the X, Y, Z values separated by commas, spaces, or tabs. There should be one point per line.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::InsertCurveFileBegin Method](#)
[IModelDoc2::InsertCurveFileEnd Method](#)
[IModelDoc2::InsertCurveFilePoint Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertCurveFileBegin Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertCurveFileBegin Method (IModelDoc2)

Creates a curve.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertCurveFileBegin()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.InsertCurveFileBegin()
```

C#

```
void InsertCurveFileBegin()
```

C++/CLI

```
void InsertCurveFileBegin();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertCurveFileBegin](#).

Example

[Insert Free Point Curve Feature \(C#\)](#)
[Insert Free Point Curve Feature \(VB.NET\)](#)
[Insert Free Point Curve Feature \(VBA\)](#)

Remarks

This method:

- is the first method called that create a curve. This method must precede all calls to [IModelDoc2::InsertCurveFilePoint](#).
- and [IModelDoc2::InsertCurveFileEnd](#) must enclose all of the calls to [IModelDoc2::InsertCurveFilePoint](#).

See the examples.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::InsertCurveFile Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertCurveFileEnd Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertCurveFileEnd Method (IModelDoc2)

Creates a curve.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertCurveFileEnd() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Boolean

value = instance.InsertCurveFileEnd()
```

C#

```
System.bool InsertCurveFileEnd()
```

C++/CLI

```
System.bool InsertCurveFileEnd();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if curve is created successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertCurveFileEnd](#).

Example

[Insert Free Point Curve Feature \(C#\)](#)
[Insert Free Point Curve Feature \(VB.NET\)](#)
[Insert Free Point Curve Feature \(VBA\)](#)

Remarks

This method:

- is the last method called to create a curve.
- and [IModelDoc2::InsertCurveFileBegin](#) must enclose all of the calls to [IModelDoc2::InsertCurveFilePoint](#).

See the examples.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::InsertCurveFile Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertCurveFilePoint Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertCurveFilePoint Method (IModelDoc2)

Creates a point for a curve.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertCurveFilePoint( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Boolean  
  
value = instance.InsertCurveFilePoint(X, Y, Z)
```

C#

```
System.bool InsertCurveFilePoint(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
System.bool InsertCurveFilePoint(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[X](#)
X value for the point

[Y](#)
Y value for the point

[Z](#)
Z value for the point

Return Value

True if this call is successful, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertCurveFilePoint](#).

Example

[Insert Free Point Curve Feature \(C#\)](#)
[Insert Free Point Curve Feature \(VB.NET\)](#)

Remarks

This method:

- sets a point for a curve.
- is called multiple times after [IModelDoc2::InsertCurveFileBegin](#) and before [IModelDoc2::InsertCurveFileEnd](#).

See the examples.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::InsertCurveFile Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertCutBlend Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertCutBlend Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertCutBlend](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertCutBlend( _  
    ByVal Closed As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Closed As System.Boolean  
  
instance.InsertCutBlend(Closed)
```

C#

```
void InsertCutBlend(  
    System.bool Closed  
)
```

C++/CLI

```
void InsertCutBlend(  
& System.bool Closed  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Closed

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertCutBlend](#).

See Also

InsertCutBlend2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertCutBlend2 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertCutBlend](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertCutBlend2( _  
    ByVal Closed As System.Boolean, _  
    ByVal KeepTangency As System.Boolean, _  
    ByVal ForceNonRational As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Closed As System.Boolean  
Dim KeepTangency As System.Boolean  
Dim ForceNonRational As System.Boolean  
  
instance.InsertCutBlend2(Closed, KeepTangency, ForceNonRational)
```

C#

```
void InsertCutBlend2(  
    System.bool Closed,  
    System.bool KeepTangency,  
    System.bool ForceNonRational  
)
```

C++/CLI

```
void InsertCutBlend2(  
& System.bool Closed,  
& System.bool KeepTangency,  
& System.bool ForceNonRational  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Closed

KeepTangency

ForceNonRational

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertCutBlend2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

InsertCutBlend3 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertCutBlend3 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertCutBlend](#).

Visual Basic (Declaration)

```
Sub InsertCutBlend3( _
    ByVal Closed As System.Boolean, _
    ByVal KeepTangency As System.Boolean, _
    ByVal ForceNonRational As System.Boolean, _
    ByVal TessToleranceFactor As System.Double, _
    ByVal StartMatchingType As System.Short, _
    ByVal EndMatchingType As System.Short _
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim Closed As System.Boolean
Dim KeepTangency As System.Boolean
Dim ForceNonRational As System.Boolean
Dim TessToleranceFactor As System.Double
Dim StartMatchingType As System.Short
Dim EndMatchingType As System.Short

instance.InsertCutBlend3(Closed, KeepTangency, ForceNonRational, TessToleranceFactor,
    StartMatchingType, EndMatchingType)
```

C#

```
void InsertCutBlend3(
    System.bool Closed,
    System.bool KeepTangency,
    System.bool ForceNonRational,
    System.double TessToleranceFactor,
    System.short StartMatchingType,
    System.short EndMatchingType
)
```

C++/CLI

```
void InsertCutBlend3(
    & System.bool Closed,
    & System.bool KeepTangency,
    & System.bool ForceNonRational,
    & System.double TessToleranceFactor,
    & System.short StartMatchingType,
    & System.short EndMatchingType
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Closed

KeepTangency

ForceNonRational

TessToleranceFactor

StartMatchingType

EndMatchingType

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertCutBlend3](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

InsertCutBlend4 Method (IModelDoc2)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertCutBlend4 Method (IModelDoc2)**Obsolete.** Superseded by [IFeatureManager.InsertCutBlend](#).

.NET Syntax

Visual Basic (Declaration)

```

Sub InsertCutBlend4( _
    ByVal Closed As System.Boolean, _
    ByVal KeepTangency As System.Boolean, _
    ByVal ForceNonRational As System.Boolean, _
    ByVal TessToleranceFactor As System.Double, _
    ByVal StartMatchingType As System.Short, _
    ByVal EndMatchingType As System.Short, _
    ByVal IsThinBody As System.Boolean, _
    ByVal Thickness1 As System.Double, _
    ByVal Thickness2 As System.Double, _
    ByVal ThinType As System.Short _
)

```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim Closed As System.Boolean
Dim KeepTangency As System.Boolean
Dim ForceNonRational As System.Boolean
Dim TessToleranceFactor As System.Double
Dim StartMatchingType As System.Short
Dim EndMatchingType As System.Short
Dim IsThinBody As System.Boolean
Dim Thickness1 As System.Double
Dim Thickness2 As System.Double
Dim ThinType As System.Short

instance.InsertCutBlend4(Closed, KeepTangency, ForceNonRational, TessToleranceFactor,
StartMatchingType, EndMatchingType, IsThinBody, Thickness1, Thickness2, ThinType)

```

C#

```

void InsertCutBlend4(
    System.bool Closed,
    System.bool KeepTangency,
    System.bool ForceNonRational,
    System.double TessToleranceFactor,
    System.short StartMatchingType,
    System.short EndMatchingType,
    System.bool IsThinBody,
    System.double Thickness1,
    System.double Thickness2,
    System.short ThinType
)

```

C++/CLI

```

void InsertCutBlend4(
    & System.bool Closed,
    & System.bool KeepTangency,
    & System.bool ForceNonRational,
    & System.double TessToleranceFactor,
    & System.short StartMatchingType,
    & System.short EndMatchingType,
    & System.bool IsThinBody,
    & System.double Thickness1,
    & System.double Thickness2,

```

```
& System.short ThinType
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Closed

KeepTangency

ForceNonRational

TessToleranceFactor

StartMatchingType

EndMatchingType

IsThinBody

Thickness1

Thickness2

ThinType

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertCutBlend4](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertCutSurface Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertCutSurface Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertCutSurface](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertCutSurface( _
    ByVal Flip As System.Boolean, _
    ByVal KeepPieceIndex As System.Integer _
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim Flip As System.Boolean
Dim KeepPieceIndex As System.Integer

instance.InsertCutSurface(Flip, KeepPieceIndex)
```

C#

```
void InsertCutSurface(
    System.bool Flip,
    System.int KeepPieceIndex
)
```

C++/CLI

```
void InsertCutSurface(
    & System.bool Flip,
    & System.int KeepPieceIndex
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Flip

True to flip the direction, false to not

KeepPieceIndex

Piece to keep if there is ambiguity

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertCutSurface](#).

Remarks

When there is ambiguity in the result of a cut, KeepPieceIndex is used to resolve which of the possible results is used. This can be set to -1 if there is no ambiguity; otherwise, it should be the index of the result, starting from 0 (up to one less than the possible number of outcomes).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[ISurfaceCutFeatureData Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertCutSwept Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertCutSwept Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertCutSwept3](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertCutSwept( _  
    ByVal Propagate As System.Boolean, _  
    ByVal Alignment As System.Boolean, _  
    ByVal KeepNormalConstant As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Propagate As System.Boolean  
Dim Alignment As System.Boolean  
Dim KeepNormalConstant As System.Boolean  
  
instance.InsertCutSwept(Propagate, Alignment, KeepNormalConstant)
```

C#

```
void InsertCutSwept(  
    System.bool Propagate,  
    System.bool Alignment,  
    System.bool KeepNormalConstant  
)
```

C++/CLI

```
void InsertCutSwept(
& System.bool Propagate,
& System.bool Alignment,
& System.bool KeepNormalConstant
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Propagate

Alignment

KeepNormalConstant

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertCutSwept](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertCutSwept2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertCutSwept2 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertCutSwept3](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertCutSwept2( _
    ByVal Propagate As System.Boolean, _
    ByVal Alignment As System.Boolean, _
    ByVal TwistCtrlOption As System.Short, _
    ByVal KeepTangency As System.Boolean, _
    ByVal ForceNonRational As System.Boolean _
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim Propagate As System.Boolean
Dim Alignment As System.Boolean
Dim TwistCtrlOption As System.Short
Dim KeepTangency As System.Boolean
Dim ForceNonRational As System.Boolean
```

```
instance.InsertCutSwept2(Propagate, Alignment, TwistCtrlOption, KeepTangency, ForceNonRational)
```

C#

```
void InsertCutSwept2(
    System.bool Propagate,
    System.bool Alignment,
    System.short TwistCtrlOption,
    System.bool KeepTangency,
    System.bool ForceNonRational
)
```

C++/CLI

```
void InsertCutSwept2(
& System.bool Propagate,
```

```
& System.bool Alignment,  
& System.short TwistCtrlOption,  
& System.bool KeepTangency,  
& System.bool ForceNonRational  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Propagate

Alignment

TwistCtrlOption

KeepTangency

ForceNonRational

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertCutSwept2](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertCutSwept3 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertCutSwept3 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertCutSwept3](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertCutSwept3(  
    ByVal Propagate As System.Boolean, _  
    ByVal Alignment As System.Boolean, _  
    ByVal TwistCtrlOption As System.Short, _  
    ByVal KeepTangency As System.Boolean, _  
    ByVal ForceNonRational As System.Boolean, _  
    ByVal StartMatchingType As System.Short, _  
    ByVal EndMatchingType As System.Short _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Propagate As System.Boolean  
Dim Alignment As System.Boolean  
Dim TwistCtrlOption As System.Short  
Dim KeepTangency As System.Boolean  
Dim ForceNonRational As System.Boolean  
Dim StartMatchingType As System.Short  
Dim EndMatchingType As System.Short
```

```
instance.InsertCutSwept3(Propagate, Alignment, TwistCtrlOption, KeepTangency, ForceNonRational,  
StartMatchingType, EndMatchingType)
```

C#

```
void InsertCutSwept3(  
    System.bool Propagate,  
    System.bool Alignment,  
    System.short TwistCtrlOption,  
    System.bool KeepTangency,
```

```
System.bool ForceNonRational,  
System.short StartMatchingType,  
System.short EndMatchingType  
)
```

C++/CLI

```
void InsertCutSwept3(  
& System.bool Propagate,  
& System.bool Alignment,  
& System.short TwistCtrlOption,  
& System.bool KeepTangency,  
& System.bool ForceNonRational,  
& System.short StartMatchingType,  
& System.short EndMatchingType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Propagate

Alignment

TwistCtrlOption

KeepTangency

ForceNonRational

StartMatchingType

EndMatchingType

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertCutSwept3](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

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InsertCutSwept4 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertCutSwept4 Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertCutSwept3](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertCutSwept4( _  
    ByVal Propagate As System.Boolean, _  
    ByVal Alignment As System.Boolean, _  
    ByVal TwistCtrlOption As System.Short, _  
    ByVal KeepTangency As System.Boolean, _  
    ByVal ForceNonRational As System.Boolean, _  
    ByVal StartMatchingType As System.Short, _  
    ByVal EndMatchingType As System.Short, _  
    ByVal IsThinBody As System.Boolean, _  
    ByVal Thickness1 As System.Double, _  
    ByVal Thickness2 As System.Double, _  
    ByVal ThinType As System.Short _  
)
```

Visual Basic (Usage)

 Copy Code

```

Dim instance As IModelDoc2
Dim Propagate As System.Boolean
Dim Alignment As System.Boolean
Dim TwistCtrlOption As System.Short
Dim KeepTangency As System.Boolean
Dim ForceNonRational As System.Boolean
Dim StartMatchingType As System.Short
Dim EndMatchingType As System.Short
Dim IsThinBody As System.Boolean
Dim Thickness1 As System.Double
Dim Thickness2 As System.Double
Dim ThinType As System.Short

```

```

instance.InsertCutSwept4(Propagate, Alignment, TwistCtrlOption, KeepTangency, ForceNonRational,
StartMatchingType, EndMatchingType, IsThinBody, Thickness1, Thickness2, ThinType)

```

C#

```

void InsertCutSwept4(
    System.bool Propagate,
    System.bool Alignment,
    System.short TwistCtrlOption,
    System.bool KeepTangency,
    System.bool ForceNonRational,
    System.short StartMatchingType,
    System.short EndMatchingType,
    System.bool IsThinBody,
    System.double Thickness1,
    System.double Thickness2,
    System.short ThinType
)

```

C++/CLI

```

void InsertCutSwept4(
    & System.bool Propagate,
    & System.bool Alignment,
    & System.short TwistCtrlOption,
    & System.bool KeepTangency,
    & System.bool ForceNonRational,
    & System.short StartMatchingType,
    & System.short EndMatchingType,
    & System.bool IsThinBody,
    & System.double Thickness1,
    & System.double Thickness2,
    & System.short ThinType
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Propagate

Alignment

TwistCtrlOption

KeepTangency

ForceNonRational

StartMatchingType

EndMatchingType

IsThinBody

Thickness1

Thickness2

ThinType

See [ModelDoc2::InsertCutSwept4](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertDatumTag2 Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertDatumTag2 Method (IModelDoc2)

Inserts a datum tag symbol at a selected location.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertDatumTag2() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Object

value = instance.InsertDatumTag2()
```

C#

```
System.object InsertDatumTag2()
```

C++/CLI

```
System.Object^ InsertDatumTag2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Newly created [datum tag](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertDatumTag2](#).

Example

[Get Annotations Arrays \(C#\)](#)
[Get Annotations Arrays \(VB.NET\)](#)
[Get Annotations Arrays \(VBA\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::InsertDatumTag2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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InsertDatumTargetSymbol Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertDatumTargetSymbol Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::InsertDatumTargetSymbol2](#).

Visual Basic (Declaration)

```
Function InsertDatumTargetSymbol( _
    ByVal Datum1 As System.String, _
    ByVal Datum2 As System.String, _
    ByVal Datum3 As System.String, _
    ByVal AreaStyle As System.Short, _
    ByVal AreaOutside As System.Boolean, _
    ByVal Value1 As System.Double, _
    ByVal Value2 As System.Double, _
    ByVal ValueStr1 As System.String, _
    ByVal ValueStr2 As System.String, _
    ByVal ArrowsSmart As System.Boolean, _
    ByVal ArrowStyle As System.Short, _
    ByVal LeaderLineStyle As System.Short, _
    ByVal LeaderBent As System.Boolean, _
    ByVal ShowArea As System.Boolean, _
    ByVal ShowSymbol As System.Boolean _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim Datum1 As System.String
Dim Datum2 As System.String
Dim Datum3 As System.String
Dim AreaStyle As System.Short
Dim AreaOutside As System.Boolean
Dim Value1 As System.Double
Dim Value2 As System.Double
Dim ValueStr1 As System.String
Dim ValueStr2 As System.String
Dim ArrowsSmart As System.Boolean
Dim ArrowStyle As System.Short
Dim LeaderLineStyle As System.Short
Dim LeaderBent As System.Boolean
Dim ShowArea As System.Boolean
Dim ShowSymbol As System.Boolean
Dim value As System.Boolean

value = instance.InsertDatumTargetSymbol(Datum1, Datum2, Datum3, AreaStyle, AreaOutside, Value1,
Value2, ValueStr1, ValueStr2, ArrowsSmart, ArrowStyle, LeaderLineStyle, LeaderBent, ShowArea,
ShowSymbol)
```

C#

```
System.bool InsertDatumTargetSymbol(
    System.string Datum1,
    System.string Datum2,
    System.string Datum3,
    System.short AreaStyle,
    System.bool AreaOutside,
    System.double Value1,
    System.double Value2,
    System.string ValueStr1,
    System.string ValueStr2,
    System.bool ArrowsSmart,
    System.short ArrowStyle,
    System.short LeaderLineStyle,
    System.bool LeaderBent,
    System.bool ShowArea,
    System.bool ShowSymbol
)
```

C++/CLI

```

System.bool InsertDatumTargetSymbol(
& System.String^ Datum1,
& System.String^ Datum2,
& System.String^ Datum3,
& System.short AreaStyle,
& System.bool AreaOutside,
& System.double Value1,
& System.double Value2,
& System.String^ ValueStr1,
& System.String^ ValueStr2,
& System.bool ArrowsSmart,
& System.short ArrowStyle,
& System.short LeaderLineStyle,
& System.bool LeaderBent,
& System.bool ShowArea,
& System.bool ShowSymbol
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Datum1

Datum2

Datum3

AreaStyle

AreaOutside

Value1

Value2

ValueStr1

ValueStr2

ArrowsSmart

ArrowStyle

LeaderLineStyle

LeaderBent

ShowArea

ShowSymbol

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertDatumTargetSymbol](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertDeleteFace Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertDeleteFace Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::InsertDeleteFace2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertDeleteFace()
```

Visual Basic (Usage)

Copy Code

```
Dim instance As IModelDoc2  
instance.InsertDeleteFace()
```

C#

```
void InsertDeleteFace()
```

C++/CLI

```
void InsertDeleteFace();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertDeleteFace](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertDeleteFace2 Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertDeleteFace2 Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::InsertDeleteFace](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertDeleteFace2( _  
    ByVal Refill As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Refill As System.Integer  
Dim value As System.Boolean  
  
value = instance.InsertDeleteFace2(Refill)
```

C#

```
System.bool InsertDeleteFace2(  
    System.int Refill  
)
```

C++/CLI

```
System.bool InsertDeleteFace2(  
& System.int Refill  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Refill

- 1 refills the face after it is deleted
- 0 does not

Return Value

True if a [DeleteFace feature](#) is inserted in the model, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertDeleteFace2](#).

Example

[Delete Selected Faces \(VBA\)](#)

Remarks

This is a part-level operation and only works when the model is a [IPartDoc](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IFeatureManager::EditDeleteFace Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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InsertDeleteHole Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertDeleteHole Method (IModelDoc2)

Obsolete. Superseded by [IFeatureManager::InsertDeleteHoleForSurface](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertDeleteHole()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.InsertDeleteHole()
```

C#

```
void InsertDeleteHole()
```

C++/CLI

```
void InsertDeleteHole();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertDeleteHole](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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InsertDome Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : InsertDome Method (IModelDoc2)

Inserts a dome.

Visual Basic Syntax

Visual Basic (Declaration)

```
Sub InsertDome( _  
    ByVal Height As System.Double, _  
    ByVal ReverseDir As System.Boolean, _  
    ByVal DoEllipticSurface As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Height As System.Double  
Dim ReverseDir As System.Boolean  
Dim DoEllipticSurface As System.Boolean  
  
instance.InsertDome(Height, ReverseDir, DoEllipticSurface)
```

C#

```
void InsertDome(  
    System.double Height,  
    System.bool ReverseDir,  
    System.bool DoEllipticSurface  
)
```

C++/CLI

```
void InsertDome(  
& System.double Height,  
& System.bool ReverseDir,  
& System.bool DoEllipticSurface  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Height
Height for the dome in meters
ReverseDir
True if you want to reverse the dome direction, false if not
DoEllipticSurface
RUE if you want the dome to be an elliptical surface, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::InsertDome](#).

Example

[Create and Modify Dome Feature \(C#\)](#)
[Create and Modify Dome Feature \(VB.NET\)](#)
[Create and Modify Dome Feature \(VBA\)](#)

Remarks

Before using this method, you must select the face for the dome using a selection mark of 1. See [IModelDocExtension::SelectByID2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0